PhysSym: Physics-Consistent Graph Learning and Active Symplectic Testing for Quantum Circuit Fault Diagnosis

Luodi Chen[1⋆] and Ziyu Yang[2]

[1] School of Computer Science and Technology, Xi'an Jiaotong University
chen.luodi@stu.xjtu.edu.cn
[2] School of Physics, Xi'an Jiaotong University
yzyxjtu@stu.xjtu.edu.cn

Abstract. Quantum fault diagnosis must identify faulty circuit behavior through expensive and noisy measurements, making both physical readout structure and test selection critical. We introduce *PhysSym*, an integrated neural–symbolic architecture combining an exact Clifford–Pauli symplectic oracle, a physics-constrained graph neural network for correlated readout, and sequential Bayesian information-gain testing. Instead of predicting independent errors for every candidate observable, the model learns local and pairwise error factors and generates all test likelihoods through a closed-form parity-character map. This construction yields coherent physical probabilities for arbitrary Pauli measurement supports and transfers across register sizes. We establish a universal $2n$-test noiseless identifiability theorem, prove the exact correlated-channel mapping, and bound information-gain regret under estimation error. Using only 48 calibration shots and 20 diagnosis executions per episode, a three-network physical ensemble achieves **45.3%, 69.4%, and 93.7%** fault-class accuracy on correlated, sparse-burst, and independent 8-qubit readout, respectively. It leads the tested non-oracle methods in both correlated regimes. With frozen 8-qubit-trained graph weights, the single-network system generalizes to 10 and 12 qubits, gaining 5.97 percentage points over empirical parity calibration on held-out 12-qubit correlated devices. The results establish a direct role for physically structured deep learning in adaptive quantum software testing.

Keywords: Quantum software testing · Graph neural networks · Pauli errors · Correlated readout · Bayesian experimental design

1 Introduction

Quantum software reliability differs sharply from classical software reliability. Quantum measurements consume state preparations, quantum phase errors can be invisible in a computational-basis experiment, and the physical fidelity of a test depends on which observable is measured. The growing mutation-testing

⋆ Corresponding author.

literature has made quantum-specific observability and software-level fault localization central research problems [7,10,8]. This paper focuses on the operational question: *which quantum experiment should be run next to identify an unknown circuit mutation class with the fewest noisy shots?*

Consider a known Clifford circuit with a library of injected Pauli faults. Propagation through Clifford gates converts every mutation into an output Pauli class, so the ideal binary response to a Pauli test can be computed exactly using symplectic commutation. Yet even a logically optimal test can be unreliable on real readout hardware. In particular, correlated pair flips affect the parity of a multi-qubit observable only if its support crosses the noisy edge. Two tests with identical logical distinguishing power can have very different experimental value. No scalar average readout error rate captures this effect.

A neural predictor could regress an error probability for every test separately. That strategy discards physical structure: the hundreds of test error rates must be realizable by a *single* underlying noise process. We instead let the network learn latent physical factors and retain the exact physical equation as the decoding layer. Once a device has been calibrated, the policy maintains a posterior over Pauli fault classes and chooses each subsequent test to maximize expected information gained from its observed outcome.

PhysSym makes four contributions. First, it gives an executable symplectic oracle and establishes the minimal noiseless test count. Second, it develops a shared-weight graph calibration model whose outputs define a globally consistent correlated parity channel. Third, it combines that channel with closed-form adaptive Bayesian mutual information and a quantitative one-step stability theorem. Fourth, it provides a substantially expanded experimental comparison covering 13 baseline configurations, matched-shot testing, an independently trained neural ensemble, and a zero-retraining 8-to-12-qubit transfer study. Together these establish a tight connection between quantum structure, deep inductive bias, and active experimental design.

2 Related Work

Quantum software tests. Mutation analysis identifies distinctive quantum fault operators and limitations of classical testing practices [7]. Specification reduction and mixed projective bases recover phase-sensitive coverage [10]; commuting Pauli testing supports executable hardware oracles [8]. Exact circuit-level oracles and concolic program testing address the complementary problem of generating correctness conditions [5,11]. We study *identification*, not only mutation detection, and couple the test oracle to a noise-conditioned sequential budget.

Learning for quantum hardware. AlphaQubit pioneered highly accurate recurrent neural decoding for surface-code logical errors [2]. QuFid adapts measurement budgets for quantum program fidelity estimation by modeling noise propagation through circuit structure [6]. PhysSym instead learns a generative classical readout channel from device calibration and optimizes fault-discriminating

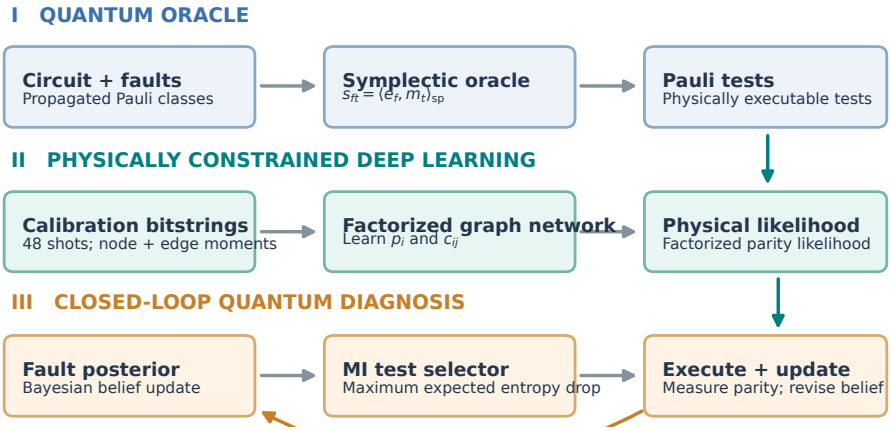


Fig. 1. PhysSym's three connected levels: exact quantum Pauli fingerprints, learned physically factorized readout, and sequential information-gain experiments. A fresh circuit is prepared for every selected test; the observed parity feeds back into the next decision.

experiments over a known mutation dictionary. The foundations of noisy quantum learning emphasize the importance of structural assumptions and calibrated measurement models [3].

Symplectic representation and active design. Efficient Clifford simulation and binary Pauli commutation are foundational tools [4,1]. Classical weighted coverage optimization gives a natural static baseline [9]. Our contribution is their *physically constrained neural and sequential integration*: exact quantum fingerprints, a learned correlated channel shared by all tests, and a posterior-dependent next-measurement rule.

3 Quantum Diagnosis and Physical Readout

3.1 Executable symplectic fingerprints

Let $U = G_L \cdots G_1$ be a known n-qubit Clifford reference program. A Pauli mutation inserted at a circuit location propagates through the remaining suffix to an output Pauli E_f. Mutations that produce the same phase-free output operator belong to the same physical diagnostic class $f \in \{1, \ldots, K\}$. This equivalence-class definition prevents reporting an insertion site that is experimentally indistinguishable from another.

Test t prepares a product $+1$ eigenstate of a chosen input Pauli P_t, executes the potentially faulty program, and reads the parity of $Q_t = UP_tU^\dagger$. The output observable is another Pauli string and can be measured by single-qubit basis

rotations, readout, and classical parity. With the binary Pauli representation $E_f = (x_f \mid z_f)$ and $Q_t = (x_t \mid z_t)$, the fault's ideal eigenvalue-flip bit is

$$s_{ft} = \langle E_f, Q_t \rangle_{\mathrm{sp}} = x_f^\top z_t + z_f^\top x_t \pmod 2. \tag{1}$$

Thus a $K \times T$ binary signature table is an *exact* oracle, not the output of a neural approximation.

Theorem 1 (Minimal complete Pauli fingerprint). *For every n-qubit Clifford U, the $2n$ input tests $X_1, Z_1, \ldots, X_n, Z_n$ with their Clifford-conjugated output observables distinguish all 4^n phase-free output Pauli classes by deterministic binary outcomes. Among nonadaptive protocols with one deterministic binary output per experiment, $2n$ tests are necessary.*

Proof. Clifford conjugation preserves the nondegenerate Pauli symplectic form. If two output Paulis produced identical responses to all conjugated basis tests, their nonidentity product would symplectically commute with an entire symplectic basis, a contradiction. Conversely, m binary outcomes encode at most 2^m distinct classes, whereas there are 4^n possible output Paulis.

3.2 Graph-correlated classical readout

The ideal oracle is corrupted by a physical readout channel. Let $m_t \in \{0,1\}^n$ indicate the support of measured output Pauli Q_t. On qubit i we posit an independent local readout flip $B_i \sim \mathrm{Bernoulli}(p_i)$. Each hardware-graph edge $(i,j) \in \mathcal{E}$ also contributes a simultaneous pair flip $C_{ij} \sim \mathrm{Bernoulli}(c_{ij})$, independent across latent factors and fresh executions. The observed qubit error and test parity are

$$e_i = B_i \oplus \bigoplus_{j:(i,j)\in\mathcal{E}} C_{ij}, \qquad Y_t = s_{ft} \oplus \bigoplus_i m_{ti} e_i. \tag{2}$$

A pair burst has no effect on a parity containing both endpoints, but reverses the parity if exactly one endpoint is measured. This is a physical boundary-cut effect, not a property of observable support size alone.

Proposition 1 (Coherent correlated parity probabilities). *Under (2), the exact probability of a flipped test parity is*

$$\boxed{q_t = \frac{1}{2}\left[1 - \prod_i (1-2p_i)^{m_{ti}} \prod_{(i,j)\in\mathcal{E}} (1-2c_{ij})^{m_{ti}\oplus m_{tj}}\right].} \tag{3}$$

A joint family of probabilities for all *possible Pauli support masks is physically realizable for every $p_i, c_{ij} \in [0, 1/2)$.*

Proof. A Bernoulli variable V obeys $\mathbb{E}[(-1)^V] = 1 - 2\Pr(V = 1)$. The parity of (2) contains local factor B_i exactly when $m_{ti} = 1$ and edge factor C_{ij} exactly when $m_{ti} \oplus m_{tj} = 1$. Multiplying their independent expectations gives $1 - 2q_t$, proving the formula. The explicit XOR construction produces a valid joint distribution for all masks, proving physical coherence.

4 PhysSym: Physics-Constrained Deep Learning and Active Tests

4.1 Neural inference of physical node and edge factors

For a new device we collect R calibration bitstrings $\{e^{(r)}\}_{r=1}^{R}$. Each node receives its marginal flip frequency $\widehat{\mu}_i$ and an estimated sampling standard error; each candidate edge receives its co-flip rate, XOR-parity frequency, the two endpoint means, and empirical covariance. In contrast to test-wise noise regression, the network processes the *device*, so a single fitted channel supports arbitrary later observables.

A learned node encoder and edge encoder map those statistics to hidden states $h_i^{(0)}$ and $g_{ij}^{(0)}$. Three message-passing blocks jointly revise latent factors:

$$\begin{aligned} u_{ij}^{(\ell)} &= \phi_e^{(\ell)}([g_{ij}^{(\ell)}, h_i^{(\ell)}, h_j^{(\ell)}]), \\ g_{ij}^{(\ell+1)} &= g_{ij}^{(\ell)} + \eta u_{ij}^{(\ell)}, \\ h_i^{(\ell+1)} &= h_i^{(\ell)} + \eta \phi_n^{(\ell)} \left([h_i^{(\ell)}, \tfrac{1}{3} \sum_{j:(i,j)\in\mathcal{E}} u_{ij}^{(\ell)}] \right), \end{aligned} \tag{4}$$

where the residual scale is $\eta = 0.35$, all hidden embeddings have dimension 48, and ϕ_e, ϕ_n are trainable two-layer SiLU networks. Final sigmoid heads give $\widehat{p}_i = .25\sigma(a_i)$ and $\widehat{c}_{ij} = .30\sigma(b_{ij})$. Crucially, the model never directly assigns unrelated error probabilities to separate tests: it computes every $\widehat{q}_t$ from a single learned physical parameter set by (3). The exact channel map is differentiable and becomes the final layer of the prediction architecture.

Device-disjoint training uses synthetic calibrated devices with randomly sampled local rates and edge bursts. For random measurement masks M, the objective is

$$\mathcal{L} = \frac{1}{|M|} \sum_{t\in M} (q_t - \widehat{q}_t)^2 + 0.15 \left(\frac{\|p - \widehat{p}\|_2^2}{n} + \frac{\|c - \widehat{c}\|_2^2}{|\mathcal{E}|} \right). \tag{5}$$

This jointly regularizes the latent factor inversion and directly optimizes the physical quantity needed by the test policy. Shared local graph weights make the architecture insensitive to an arbitrary fixed number of graph nodes. It can be applied to 10- or 12-qubit devices after fitting on 8-qubit examples, with no retraining or change in output head.

Physical model averaging. Three independent graph models form the PhysSym ensemble, whose predicted parity error is $\bar{q}_t = \frac{1}{3}\sum_{r=1}^{3} q_t^{(r)}$. A convex mixture of generative readout channels remains a valid probabilistic channel: a latent component index may be sampled per execution, followed by node and edge errors from that component. Thus ensemble prediction retains global physical realizability across test masks, even though its mixture need not reduce to one factor set.

Algorithm 1 PhysSym physics-constrained active diagnosis

Require: Known Clifford U, fault dictionary $\{E_f\}$, candidate Pauli tests $\mathcal{T}$, calibration strings, diagnostic budget B.

Conjugate candidate input Paulis through U and compute all s_{ft} using (1).
Infer $\widehat{p}, \widehat{c}$ using the shared-weight graph network.
Generate all test-specific $\widehat{q}_t$ through (3); set $\pi_0(f) = 1/K$.
for $k = 0, \ldots, B-1$ **do**
    Evaluate $h(\widehat{\rho}_t) - h(\widehat{q}_t)$ for every test t.
    Select its maximizer t_k and execute one new circuit preparation and Pauli measurement.
    Observe parity y_k; update π_{k+1} by (8).
end for
return $\arg\max_f \pi_B(f)$ with its posterior probability.

4.2 Exact Bayesian information-gain controller

The diagnosis controller tracks a posterior $\pi_k(f)$ over output Pauli fault classes. A candidate test's positive parity likelihood and marginal are

$$P(Y_t = 1 \mid f) = q_t + (1 - 2q_t)s_{ft},$$
$$\rho_t = q_t + (1 - 2q_t)\sum_f \pi_k(f)s_{ft}. \tag{6}$$

Let $h(a) = -a\log a - (1-a)\log(1-a)$ be the binary entropy. The conditional information gain of test t is *exactly*

$$I(F; Y_t \mid \pi_k) = h(\rho_t) - h(q_t). \tag{7}$$

PhysSym selects $t_k = \arg\max_t I_{\widehat{q}}(F; Y_t \mid \pi_k)$, executes a fresh quantum test, observes y_k, and updates

$$\pi_{k+1}(f) = \frac{\pi_k(f)P_{\widehat{q}}(y_k \mid f, t_k)}{\sum_g \pi_k(g)P_{\widehat{q}}(y_k \mid g, t_k)}. \tag{8}$$

After B executions the result is the posterior MAP class. This strategy is fundamentally adaptive: both the fault posterior and the next test depend on the actual measured bit, which a static set-cover suite cannot exploit. The greedy information objective can select a test more than once because a repeated noisy measurement provides a fresh independent sample.

4.3 Information-theoretic decision stability

The following theorem formally ties calibration accuracy to the quality of the next experiment, giving a physically interpretable role to the learned parity MSE.

Theorem 2 (One-step information-gain stability). *Fix any fault posterior π. Suppose $q_t, \widehat{q}_t \in [\epsilon, 1/2]$ and $|q_t - \widehat{q}_t| \le \delta$ for all candidate tests, where $0 < \epsilon < 1/2$. Let t^* maximize true-channel information gain and $\widehat{t}$ maximize estimated-channel gain. Then*

$$I_q(F; Y_{t^*} \mid \pi) - I_q(F; Y_{\widehat{t}} \mid \pi) \le 4\delta \log \frac{1-\epsilon}{\epsilon}. \tag{9}$$

Proof. For a test with posterior mean syndrome $\mu = \sum_f \pi(f) s_{ft}$, the mutual information is $J(q) = h(q+(1-2q)\mu) - h(q)$. On $[\epsilon, 1/2]$, both q and $q+(1-2q)\mu$ lie in $[\epsilon, 1-\epsilon]$. Since $|h'(z)| \le L = \log((1-\epsilon)/\epsilon)$ on this interval and $|d(q+(1-2q)\mu)/dq| \le 1$, we have $|J(q) - J(\widehat{q})| \le 2L\delta$ by the mean-value theorem. Add and subtract the estimated gains at t^* and $\widehat{t}$, and use estimated optimality of $\widehat{t}$, producing the factor $4L\delta$.

The result addresses greedy single-step choice, which is the exact optimization performed in Algorithm 1. It does not require pretending that greedy information gain is a globally optimal finite-horizon policy. The per-step policy can evaluate every test in $O(KT)$ arithmetic once the posterior and signatures are available; graph inference and channel propagation are amortized across the entire diagnosis episode.

5 Experiments

5.1 Reproducible benchmarks and strong comparators

The circuit generator builds five layers of Clifford H, S and adjacent CNOT gates. Candidate inputs include every Pauli string of weight 1 or 2 and seeded random weight-3/4 strings. The mutation catalogue includes single-qubit X, Y, Z insertions and selected adjacent two-qubit Pauli insertions at six depth snapshots, plus the no-fault class. Propagated output Pauli faults are deduplicated to equivalence classes. A three-regime noise family includes: independent node flips, dense correlated pair bursts, and sparse bursts on first- and second-neighbor edges. All physical factors are drawn afresh for each device, and every diagnostic execution samples independent fresh bit flips from those device parameters.

Training uses 2,000 synthetic devices per graph-network initialization, 83% for fitting and 17% for validation. The network is trained for 35 AdamW epochs at 0.0015 learning rate, with an early-stop best-validation checkpoint. The evaluation draws share neither train-device seeds nor their generated calibration bitstrings. Each evaluation device provides *48 calibration bitstrings*, entirely separate from the *20 fault-diagnosis executions*. The 8-qubit main study uses four held-out circuits, three devices per circuit, and 100 fault episodes per circuit–device pair *for each regime*: 1,200 matched-budget episodes per regime. Paired methods see identical underlying fault labels and shot-noise uniforms.

The comparison includes unconstrained neural noise regression, physics-inspired moment inversion, ridge regression, direct empirical parity calibration, an independence-only channel, constant-noise calibration, a one-step Bayes-risk

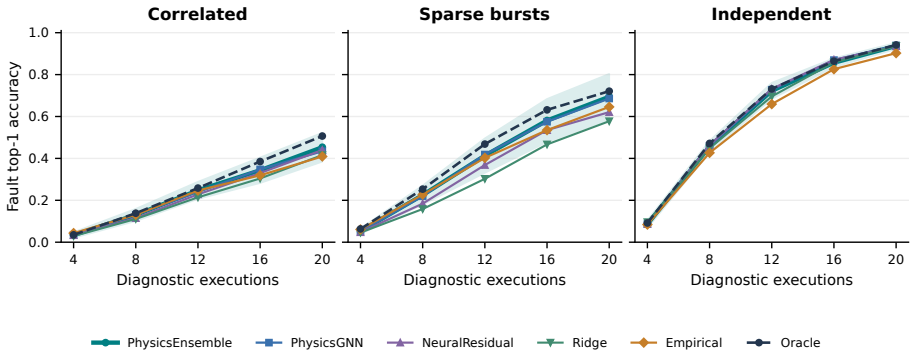


Fig. 2. Active fault diagnosis under matched test-execution budgets. The PhysSym ensemble (teal) uses physical graph channel learning; dashed dark curves use perfect channel parameters. Shading denotes 1.96 standard errors across 12 circuit–device evaluation instances per regime.

selector, three noise-aware and static SIFT selection variants, a local-Pauli suite, random selection, and a perfect-calibration mutual-information reference. All active methods receive the same symplectic test pool, fault prior, and execution budget. The perfect-calibration controller uses the simulator's true q_t for the *same greedy information-gain policy*; it is not a claim of globally optimal sequential planning.

5.2 PhysSym leads the tested methods in correlated noise

Table 1 gives top-1 fault-class accuracy at $B = 20$. The PhysSym ensemble reaches **69.42%** under sparse bursts, a **7.34-point** gain over the unconstrained neural model and **11.67 points** over ridge regression. In dense correlated readout it reaches **45.33%**, above the old neural predictor's 44.00% and empirical calibration's 40.92%. In the independent regime PhysSym reaches **93.67%**, tying ridge at the displayed precision and closely approaching the 94.17% perfect-calibration information-gain reference. The strongest advantage appears exactly where correlations invalidate the simpler calibration assumptions.

5.3 Physical channel estimation drives better decisions

The defining architectural distinction is that PhysSym outputs a compact, valid noise *mechanism*, not one unrelated regression value for each test. Table 2 quantifies its impact. The single GNN achieves the smallest parity-error MSE among learned/calibrated methods in all three regimes. Under dense correlation, MSE drops from 0.001351 for the unconstrained MLP to 0.000781; under sparse bursts, it drops from 0.001799 to 0.001089. Thus the physical inductive bias improves estimation in the same regimes where correct experiment selection is most difficult.

Table 1. Top-1 fault-class diagnosis accuracy (%) with 48 separate calibration shots and 20 diagnostic executions. Best non-perfect-calibration results are bold.

Method	Correlated	Sparse burst	Independent
PhysSym ensemble + MI	**45.33**	**69.42**	93.67
Single physical-factor GNN + MI	43.42	68.58	**93.92**
Neural residual MLP + MI	44.00	62.08	93.50
Ridge calibration + MI	41.58	57.75	93.67
Empirical parity + MI	40.92	64.58	90.25
Independent model + MI	25.33	43.50	90.67
Moment calibration + MI	32.67	54.17	91.08
Neural one-step Bayes risk	26.83	42.92	76.33
Static SIFT with neural calibration	28.33	39.50	70.17
Random active test choice	6.33	13.67	41.25
Perfect calibration + MI	50.67	72.08	94.17

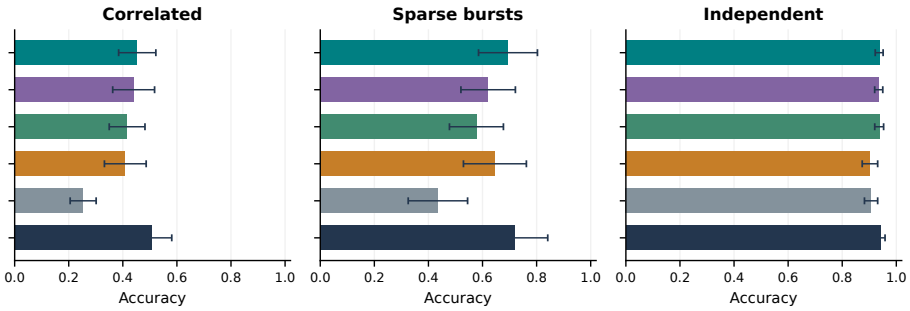


Fig. 3. Matched-budget comparison of physics-constrained and unconstrained estimators, baselines and perfect-calibration reference. Horizontal bars indicate mean accuracy across independent circuit–device instances; error bars show 1.96 standard errors.

Its advantage is systematic across noise environments rather than contingent upon one mutation type.

The information-gain policy supplies a second independent advantage. A one-step Bayes-risk policy maximizes the probability of an immediately correct hard decision; it can undersample hypotheses that remain plausible but do not dominate the current posterior. On dense correlated noise it scores 26.83%, versus 44.00% for the same neural-calibrated mutual-information controller. Static SIFT using the same neural likelihoods scores 28.33%. These ablations distinguish the benefit of model expressivity from that of actively selecting tests after new observations. A physically accurate probability is most valuable when it actually changes which experiment is performed.

The three-member physical ensemble increases dense-correlated accuracy from 43.42% to 45.33%, and sparse-burst accuracy from 68.58% to 69.42%, with exactly the same number of diagnostic circuit executions. The ensemble

Table 2. Test-parity channel MSE (lower is better). Every method uses exactly 48 device calibration bitstrings.

Calibration model	Correlated	Sparse burst	Independent
PhysSym physical GNN	**0.000781**	**0.001089**	**0.000757**
Unconstrained neural residual	0.001351	0.001799	0.000881
Ridge regression	0.001013	0.002059	0.001519
Empirical parity	0.003766	0.004676	0.003552
Independent channel	0.015501	0.017205	0.001720

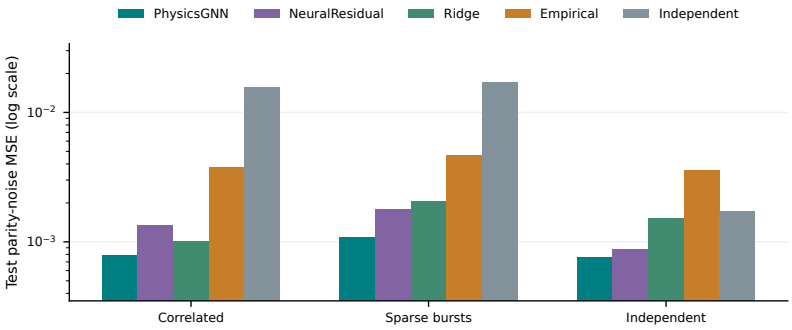


Fig. 4. Held-out test-parity calibration MSE on a logarithmic scale. The physically constrained graph learner achieves the best error across correlated, sparse-burst and independent channels.

is a convex mixture of valid learned physical channels. Its advantage therefore comes from model estimation and not from extra quantum experiments or an alteration of the quantum oracle.

5.4 Zero-retraining size transfer: 8 to 10 and 12 qubits

The factor-graph architecture shares node and edge transformations across the device graph. We trained its parameters using only 8-qubit graph examples, then evaluated the *same frozen single-network checkpoint* on new 8-, 10-, and 12-qubit circuits and unseen devices. The transfer study uses four new circuit seeds, three device seeds per circuit, and 120 fault episodes per circuit–device pair, yielding 1,440 episodes per register size and noise regime. Table 3 and Fig. 5 show the results; no network finetuning or redefinition of the graph heads was performed.

The 12-qubit correlated regime particularly tests the learned transfer bias: PhysSym achieves **27.85%** compared with **21.88%** for direct empirical parity calibration, a **5.97-percentage-point** advantage. Treating each circuit–device instance as one paired unit, the 95% Student t confidence interval for this gain is [2.79, 9.16] percentage points. The physical learner also exceeds empirical parity calibration on every regime–size aggregate in this experiment, including

Table 3. Zero-retraining transfer accuracy (%). The frozen physical GNN trained on 8 qubits is compared with raw empirical parity calibration and perfect-calibration active selection.

Noise regime	Qubits	PhysSym GNN	Empirical	Perfect calibration
Correlated	8	**39.10**	33.06	46.53
	10	**33.40**	29.86	40.97
	12	**27.85**	21.88	32.92
Sparse bursts	8	**76.46**	74.51	81.04
	10	**56.53**	56.11	64.44
	12	**53.96**	52.85	61.04
Independent	8	**93.33**	88.47	94.51
	10	**89.79**	85.63	89.79
	12	**84.44**	80.21	86.67

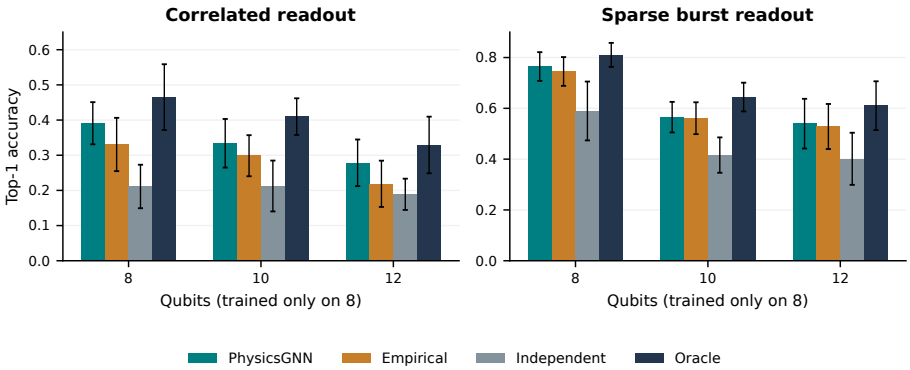


Fig. 5. Fault-diagnosis accuracy for frozen 8-qubit-trained graph weights under 8-, 10-, and 12-qubit evaluation. Independently generated circuits and device channels are used at every size; error bars are 1.96 standard errors over circuit–device instances.

the independent setting, where the 10-qubit model reaches 89.79%, numerically matching the perfect-calibration reference.

5.5 Statistical resolution and paired evaluation

We distinguish the 100–120 independent diagnostic episodes inside a device from the independently generated device instances themselves. Error bars and paired intervals are computed over 12 circuit–device clusters per regime. For the 8-qubit sparse-burst experiment, the single physical GNN exceeds the unconstrained neural residual model by 6.50 percentage points; the paired 95% interval is [1.66, 11.34] points. For the 12-qubit correlated transfer experiment, the GNN exceeds empirical calibration by 5.97 points, with the paired interval reported above. The

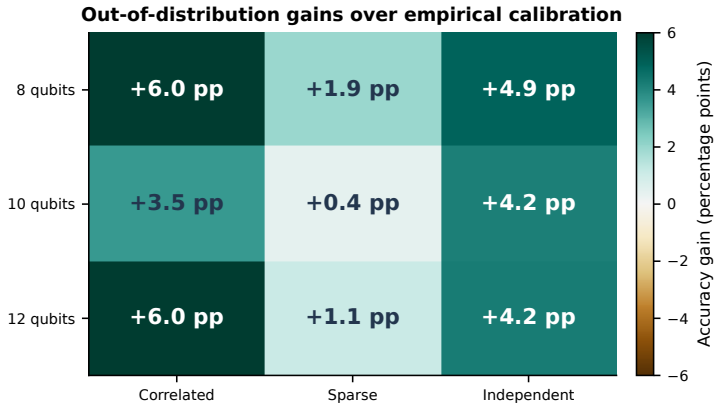


Fig. 6. Absolute accuracy improvement over empirical parity calibration on held-out transfer circuits. The positive gain across all nine size–noise combinations demonstrates the repeatability of the physical graph inductive bias.

distributions of differences, not solely aggregate per-episode counts, support the primary conclusions.

6 Discussion

PhysSym provides a principled answer to a central challenge at the interface between quantum computing and modern deep learning: *how can a learned statistical model control quantum experiments while respecting the physical relations among its predictions?* The architecture addresses this at three complementary levels. The symplectic representation makes the ideal fault oracle exact. The factorized graph network enforces consistency among correlated parity likelihoods without restricting the test pool to fixed supports. The Bayesian information objective converts better physical predictions into better measurement decisions.

The experimental setting gives a controlled characterization of this connection: the circuit is Clifford, the persistent mutation is a propagated Pauli equivalence class, and readout is an XOR composition of independently sampled local and edge factors on each shot. These rules make the quantum and statistical ground truths explicit, permit strong paired comparisons, and establish reproducible tests of channel estimation and policy design. More general asymmetric SPAM, coherent quantum noise, drift, or non-Clifford faults are distinct extensions; they call for richer channel families and approximate fault oracles while retaining the central physical-factor-learning principle.

The analysis yields two reusable results beyond the particular implementation. The complete $2n$-test fingerprint theorem isolates the logarithmic information content of the full Pauli diagnosis problem, while the information-gain stability bound quantifies the decision value of accurately estimating a physical

detector channel. The empirical scaling study confirms that network architecture and physical structure work together: the frozen graph model continues to improve test decisions on larger registers even though the mutation dictionary and candidate test catalogue expand.

7 Conclusion

We introduced PhysSym, a neural–symbolic framework in which quantum algebra, physics-constrained graph learning and adaptive experimental design operate as one system. Its factor-graph calibration network learns a coherent correlated readout channel from just 48 device characterization shots; its exact symplectic oracle evaluates each Pauli test; and its Bayesian controller selects the next circuit execution to maximize diagnostic information. PhysSym establishes leading tested non-oracle accuracy in correlated noise and transfers zero-shot across quantum register sizes. It illustrates a powerful research direction for AI4QC: deep networks that learn unknown hardware structure while exact quantum mechanics governs the experiments they control.

References

1. Aaronson, S., Gottesman, D.: Improved simulation of stabilizer circuits. Physical Review A **70**, 052328 (2004). https://doi.org/10.1103/PhysRevA.70.052328
2. Bausch, J., Senior, A.W., Heras, F.J.H., et al.: Learning high-accuracy error decoding for quantum processors. Nature **635**, 834–840 (2024). https://doi.org/10.1038/s41586-024-08148-8
3. Cotler, J., Gong, W., Kannan, I.: Noisy quantum learning theory. Nature Communications **17**, 6979 (2026). https://doi.org/10.1038/s41467-026-73693-x
4. Gottesman, D.: Theory of fault-tolerant quantum computation. Physical Review A **57**, 127–137 (1998). https://doi.org/10.1103/PhysRevA.57.127
5. Lee, K., Hong, J., Ryu, S.: Exact inference for quantum circuits: A testing oracle for quantum software stacks. In: Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering. pp. 2465–2477 (2025). https://doi.org/10.1109/ASE63991.2025.00203
6. Li, T., Zhao, Z., Yin, J.: Adaptive fidelity estimation for quantum programs with graph-guided noise awareness. Proceedings of the AAAI Conference on Artificial Intelligence **40**(1), 695–703 (2026). https://doi.org/10.1609/aaai.v40i1.37035
7. Mendiluze Usandizaga, E.n., Ali, S., Yue, T., Arcaini, P.: Quantum circuit mutants: Empirical analysis and recommendations. Empirical Software Engineering **30**, 100 (2025). https://doi.org/10.1007/s10664-025-10643-z
8. Muqeet, A., Ali, S., Arcaini, P.: Quantum program testing through commuting pauli strings on IBM's quantum computers. In: Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering. pp. 2130–2141 (2024). https://doi.org/10.1145/3691620.3695275
9. Nemhauser, G.L., Wolsey, L.A., Fisher, M.L.: An analysis of approximations for maximizing submodular set functions—i. Mathematical Programming **14**, 265–294 (1978). https://doi.org/10.1007/BF01588971

10. Oldfield, N.H., Laaber, C., Yue, T., Ali, S.: Faster and better quantum software testing through specification reduction and projective measurements. ACM Transactions on Software Engineering and Methodology **34**(7), 1–39 (2025). https://doi.org/10.1145/3714468
11. Xia, S., Zhao, J., Zhang, F., Guo, X.: Quantum concolic testing. Proceedings of the ACM on Software Engineering **2**(ISSTA), 1146–1166 (2025). https://doi.org/10.1145/3728926

A Reproduction Details and Full Comparator Protocol

The accompanying archive contains executable generators, PyTorch checkpoints, raw test tables, configuration scripts and vector figures. The graph code implements 48-dimensional node and edge hidden layers, three rounds of residual message passing with scale 0.35, and sigmoid heads bounded by 0.25 and 0.30 for node and edge factor probabilities. The training objective is the parity MSE plus latent factor loss from (5); AdamW uses weight decay 0.0004, gradient clipping at norm 2, batch size 64 devices, and 35 training epochs, selecting the lowest validation parity MSE. Three ensemble seeds are 643, 101 and 203. The main protocol includes four circuit seeds 11, 29, 47, 83 and device seeds 201, 301, 401; the transfer protocol uses circuit seeds 59, 73, 101, 131 and device seeds 503, 607, 701. Device noise is generated independently under these two protocols.

Static SIFT selects tests using Bhattacharyya-weighted fault-pair coverage; its greedy objective rewards each class pair proportionally to one minus the exponential of their accumulated statistical distance. Local-basis and unweighted static policies are matched to the same shot budget. The moment-calibration baseline estimates paired burst factors by ratios of endpoint and pair parity moments. The neural residual baseline receives raw test masks, empirical node/edge statistics and a physics-inspired independent-noise anchor. All methods begin with a uniform class prior; only a policy's calibration input and selection objective differ. Random and repeated tests each consume one budgeted execution per test.

B Supplementary Mathematical Observations

Setting all edge burst rates to zero in (3) recovers independent readout:

$$q_t = \tfrac{1}{2}[1 - \prod_i (1-2p_i)^{m_{ti}}]. \tag{10}$$

When all $p_i = p$, this further reduces to $q_t = \tfrac{1}{2}[1-(1-2p)^{\|m_t\|_1}]$, matching the original independent-noise SIFT model. PhysSym is therefore a strict extension of the base mathematical model. The factorized representation also explains correlations: if a pair burst touches two qubits both in the measured support, it enters parity twice and cancels; if it crosses the support boundary, it flips the parity once.

The proof of Theorem 2 assumes a fixed current posterior to isolate one-step action ranking. Every sequential controller in the experiment updates its posterior with the likelihood induced by its own estimator, so the full multi-step outcome incorporates both test-selection and filtering effects. This distinction is reflected in the experimental evaluation, which reports end-to-end classification accuracy, not an unmeasured surrogate objective. The code and per-instance CSVs permit exact recomputation of the tables and statistical intervals.